Rincoin 840k S6/b Consensus Change Specification

Bounded Height-Only Customized Halving Transition

Rincoin Community Forge

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1 Status

This is a **Candidate Specification — Discussion Draft** for a possible Rincoin consensus change at block height 840,000. It is independently implementable, but it is not adopted consensus and does not activate itself.

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S6/b is a provisional deterministic bounded interpretation of Customized Halving Scenario II pending clarification from the study's author. This specification fixes the rule needed for reproducible review; it does not claim that the source study supplied the terminal derivation or every consensus detail (Tokino 2026).

2 Scope

This document defines only the maximum block subsidy by height, its activation boundary, terminal-height derivation, coinbase amount validation, and fixed test vectors. It leaves chain-separation commitments, replay policy, release engineering, and activation coordination to separate specifications.

The schedule is height-only. It does not use actual issued supply, circulating supply, spendable supply, lost or burned coins, transaction volume, price, governance state, or prior underclaims as consensus inputs.

3 Units and Conventions

- One RIN is exactly `100000000` base units.
- Height intervals are half-open: $[a, b)$ includes a and excludes b.
- The subsidy returned by the rule is the maximum amount of new RIN that the coinbase transaction may create at that height.
- Transaction fees are not new issuance.
- All calculations use non-negative integers. Consensus code must not use floating-point arithmetic.

4 Historical Rule Below Height 840,000

For every height $h < 840{,}000$, nodes use the deployed Rincoin subsidy rule without modification (Rin-coin contributors 2026). With initial subsidy 5,000,000,000 base units and 210,000-block epochs:

$$S_0(h) = 5{,}000{,}000{,}000 >> \left\lfloor \frac{h}{210{,}000} \right\rfloor.$$

Consequently, scheduled issuance before height 840,000 is exactly:

$$210{,}000 \times (50 + 25 + 12.5 + 6.25) = 19{,}687{,}500 \text{ RIN}.$$

An implementation of S6/b must produce identical subsidy results and block validity below height 840,000. It must not apply the new phases retroactively.

5 Post-Activation Maximum Subsidy

For height $h \geq 840{,}000$, the maximum subsidy $S(h)$ is:

Height interval	Maximum subsidy, base units	Maximum subsidy, RIN
$840{,}000 \leq h < 2{,}100{,}000$	400000000	4.00000000
$2{,}100{,}000 \leq h < 4{,}200{,}000$	200000000	2.00000000
$4{,}200{,}000 \leq h < 6{,}300{,}000$	100000000	1.00000000
$6{,}300{,}000 \leq h < 234{,}587{,}500$	60000000	0.60000000
$h \geq 234{,}587{,}500$	0	0.00000000

Height 840,000 is the first block governed by the new rule. Height 234,587,499 is the final block with a non-zero scheduled subsidy. Height 234,587,500 is the first zero-subsidy block.

6 Terminal-Height Derivation

The terminal height is derived; it is not an independently selected input. Exact scheduled issuance before the 0.6 RIN phase is:

Component	Blocks	RIN per block	Scheduled issuance, RIN
Deployed rule below 840,000	840,000	phase-dependent	19,687,500
$[840{,}000, 2{,}100{,}000)$	1,260,000	4	5,040,000
$[2{,}100{,}000, 4{,}200{,}000)$	2,100,000	2	4,200,000
$[4{,}200{,}000, 6{,}300{,}000)$	2,100,000	1	2,100,000
Total before height 6,300,000			**31,027,500**

The exact remaining maximum scheduled issuance is:

$$168{,}000{,}000 - 31{,}027{,}500 = 136{,}972{,}500 \text{ RIN}.$$

At 0.6 RIN per block, the final phase contains exactly:

$$136{,}972{,}500/0.6 = 228{,}287{,}500 \text{ blocks.}$$

Therefore its exclusive end, and the first zero-subsidy height, is:

$$6{,}300{,}000 + 228{,}287{,}500 = 234{,}587{,}500.$$

An implementation must derive and assert this result with integer base-unit arithmetic. It must fail its build or tests if the ceiling is not exactly divisible by the final-phase subsidy after earlier scheduled issuance is deducted.

7 Maximum-Scheduled-Issuance Semantics

Maximum scheduled issuance is the sum of the maximum subsidy permitted at every height. Under this rule it is exactly `16800000000000000` base units, or `168000000.00000000 RIN`.

A miner may claim less than the maximum subsidy. Any unclaimed amount is not reissued, does not accumulate, does not extend the 0.6 RIN phase, and does not change the subsidy at any later height. Actual issued supply can therefore be lower than maximum scheduled issuance.

Consensus validation must not scan historical coinbases to calculate a supply cap. It must not consult chainstate totals, UTXO totals, circulating-supply estimates, burns, lost coins, or wallet state when calculating $S(h)$.

8 Coinbase Amount Validation

For a block at height h, let F be the sum of transaction fees in that block and let C be the total value of all coinbase outputs. Nodes reject the block if:

$$C > S(h) + F.$$

An exact claim of $S(h) + F$ is valid if every other consensus rule passes. An underclaim is valid and permanently forgoes the difference. A claim one base unit above $S(h) + F$ is invalid. At and after height 234,587,500, the coinbase may claim fees but no new subsidy.

9 Boundary and Normative Vectors

The complete normative vector artifacts are:

- `data/S6B_normative_test_vectors.csv`;
- `data/S6B_normative_test_vectors.json`;
- `data/S6B_normative_test_vectors.sha256`.

They include every activation and phase boundary, the final two non-zero blocks, the first zero-subsidy block, and the following block. They also record cumulative maximum scheduled issuance immediately before and after each block.

Vector	Height	Phase	Maximum subsidy, base units	Maximum subsidy, RIN	Cumulative before, RIN	Cumulative after, RIN	Boundary purpose
S6B-001	839999	deployed-pre-activation	625000000	6.25000000	19687493.75000000	19687500.00000000	last pre-activation block
S6B-002	840000	S6B-4-RIN	400000000	4.00000000	19687500.00000000	19687504.00000000	activation block
S6B-003	840001	S6B-4-RIN	400000000	4.00000000	19687504.00000000	19687508.00000000	second activation-phase block
S6B-004	2099999	S6B-4-RIN	400000000	4.00000000	24727496.00000000	24727500.00000000	final 4-RIN block
S6B-005	2100000	S6B-2-RIN	200000000	2.00000000	24727500.00000000	24727502.00000000	first 2-RIN block
S6B-006	2100001	S6B-2-RIN	200000000	2.00000000	24727502.00000000	24727504.00000000	second 2-RIN block
S6B-007	4199999	S6B-2-RIN	200000000	2.00000000	28927498.00000000	28927500.00000000	final 2-RIN block
S6B-008	4200000	S6B-1-RIN	100000000	1.00000000	28927500.00000000	28927501.00000000	first 1-RIN block
S6B-009	4200001	S6B-1-RIN	100000000	1.00000000	28927501.00000000	28927502.00000000	second 1-RIN block
S6B-010	6299999	S6B-1-RIN	100000000	1.00000000	31027499.00000000	31027500.00000000	final 1-RIN block

Vector	Height	Phase	Maximum subsidy, base units	Maximum subsidy, RIN	Cumulative before, RIN	Cumulative after, RIN	Boundary purpose
S6B-011	6300000	S6B-0.6-RIN	60000000	0.60000000	31027500.00000000	31027500.60000000	first 0.6 RIN block
S6B-012	6300001	S6B-0.6-RIN	60000000	0.60000000	31027500.60000000	31027501.20000000	second 0.6 RIN block
S6B-013	234587498	S6B-0.6-RIN	60000000	0.60000000	167999998.80000000	167999999.40000000	penultimate non-zero block
S6B-014	234587499	S6B-0.6-RIN	60000000	0.60000000	167999999.40000000	168000000.00000000	final non-zero block
S6B-015	234587500	S6B-zero	0	0.00000000	168000000.00000000	168000000.00000000	first zero-subsidy block
S6B-016	234587501	S6B-zero	0	0.00000000	168000000.00000000	168000000.00000000	second zero-subsidy block

Artifact	SHA-256
data/S6B_normative_test_vectors.csv	2802045fde1e8ef0cd061a7e01dded6488fca5f2ca9c5573ca9d46dfe efb70
data/S6B_normative_test_vectors.json	50c4328578e8e46adc800a80e96796bb13572315f0130691a0d01ac

The vectors are generated by `scripts/simulate_monetary_scenarios.py` and checked by the separate `scripts/verify_s6b_independently.py`, which does not import the primary subsidy implementation.

10 Compatibility and Activation Risks

At height 840,000, S6/b permits 4 RIN while unchanged deployed software permits 3.125 RIN. The validity sets overlap when a miner underclaims: a coinbase claim at or below the lower permitted amount may be valid under both subsidy rules. Subsidy difference alone therefore does not guarantee deterministic branch separation.

All nodes, miners, pools, wallets, explorers, and services that intend to follow this rule require coordinated software and activation handling. Pool coinbase construction must include fees correctly and must never treat the 168M maximum as a stateful supply counter.

11 Chain Separation and Replay

This monetary specification does not select a chain-separation commitment or a transaction-replay policy. Those are unresolved until a separate reviewed specification fixes their exact bytes, activation behavior, and test vectors. Implementers must not infer branch identity from the subsidy amount alone.

12 Reference Implementation Placeholders

The following references must be replaced with immutable identifiers before a release decision:

- source repository and commit: `[SOURCE_REPOSITORY_AND_COMMIT]`;
- subsidy implementation path and symbol: `[SUBSIDY_IMPLEMENTATION]`;
- block-validation call site: `[BLOCK_VALIDATION_CALL_SITE]`;
- unit and functional test commits: `[TEST_COMMITS]`;
- candidate binary hashes: `[CANDIDATE_BINARY_HASHES]`.

No runtime option may choose among S1, S5/b, and S6/b in a final mainnet binary. A final release must contain one reviewed immutable schedule.

References

Rin-coin contributors. 2026. *Rincoin Core: Deployed Block-Subsidy Validation.* Version b52c87778f800dc5f4e2f59c372badbc139f933f. Released March 7. https://github.com/Rin-coin/rincoin/blob/b52c87778f800dc5f4e2f59c372badbc139f933f/src/validation.cpp#L1271-L1281.

Tokino, Michiru. 2026. *On the Convergence of Regenerative Thermodynamic Security and Economic Incentives.* Version 1.6.4. Zenodo. https://doi.org/10.5281/zenodo.20674990.